

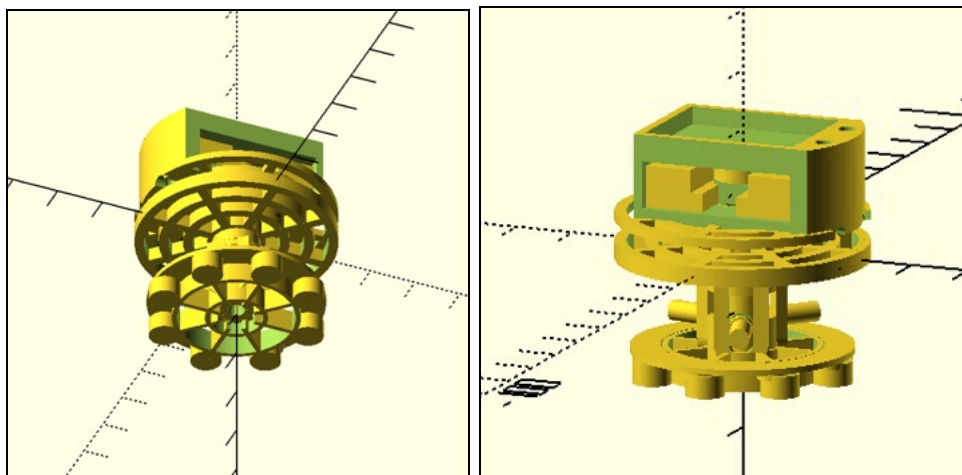
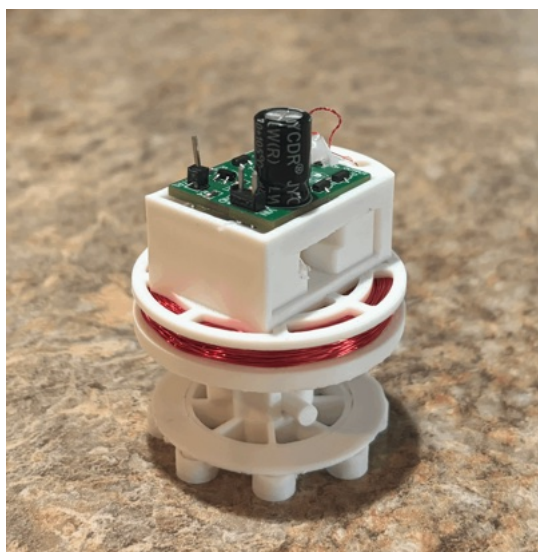


Metal Orchestra Toggle Actuator

ARAM-MOTCA_0_10-0SP*

0SPG (Pet-g) and 0SPA (PLA) variants

Spec Sheet



Product of American Robotics Assisted Manufacturing

www.ARAM-USA.com

View further kit usage information & legal information



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Overview

This mini actuator is part of the ARAM Metal Orchestra set. These are smart actuators utilizing an integrated high efficiency h-bridge, with simplified control schema requiring only a single pin. The actuator integrates a magnetic attaching base to allow these to attach to any magnetic object.

These are intended for use as part of the Metal Orchestra kit that allows users to turn household objects into an orchestra. However these have tons of additional applications, including unique I/O devices, animatronics, smart-home integration and more.

*Please read the usage guide and view legal disclaimers.



Quick Facts:

Electrical Characteristics

Continuous Voltage Range	3.3V - 5V	Over-Voltage* Range	5.1V - 12V
Average 5V Current Draw	0.35 - 0.46 Amps	H-Bridge (Driver) Resistance	0.2 ohms
Total Component Resistance	11.2 - 13.2 ohms	Solenoid Resistance	11 - 13 ohms
Input Voltage	1.8V * (Reduced Efficiency), 3.3V , and 5V		

*Actuator may run continuously at or below 5V, above this the **Over-Voltage range (5.1V - 12V)** is electrically acceptable for the driver to handle, however this will cause the armature to heat up, causing damage over time if not properly cooled.

*1.8 Volts will work for logic, However this has reduced efficiency (greater effective resistance) and will possibly reduce the working life.

Force Characteristics

	3.3V	5V
Strike Time	50 ms	20 ms
Force (Average Over Strike)	0.09 N	0.22 N

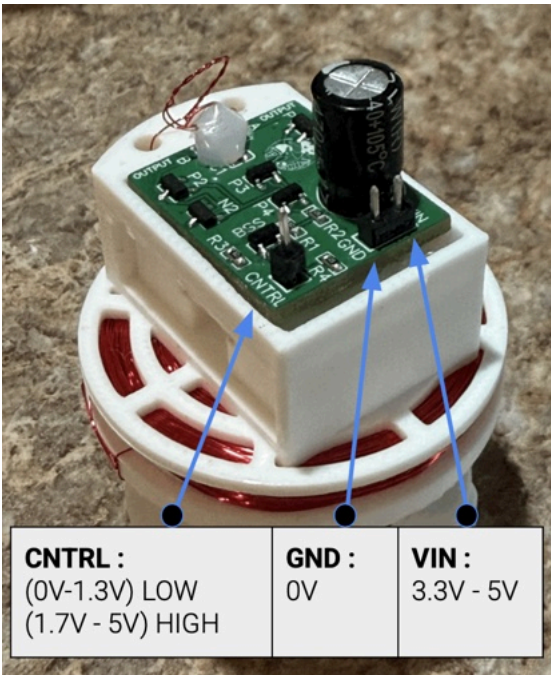
Plunger Weight	7g
Force of Gravity on Plunger	0.068 N
Travel Distance	4.5mm +/- 0.15 mm

*Force may be limited by magnetic interference

Material Options: Pet-G, PLA



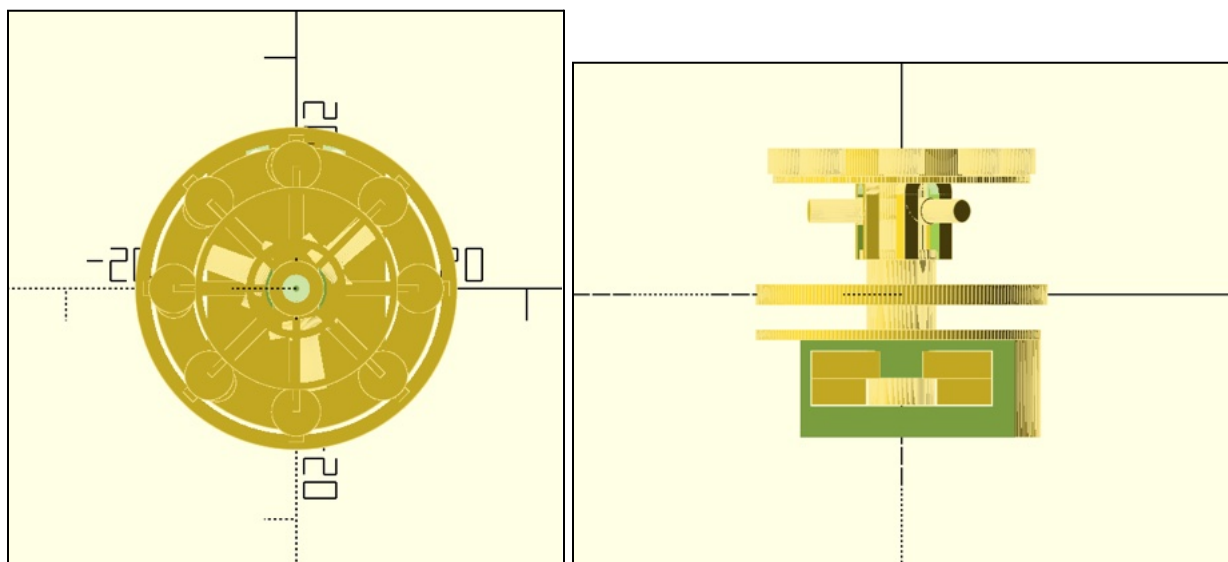
Pinout and Basic Usage



Pin	Voltage	Additional Comments
VIN	3.3V to 5V Continuous 5V+ to 12V Pulsed	Incorrectly plugging VIN to 0V may damage your unit
GND	0V	Incorrectly plugging GND to voltage may damage your unit
CNTRL	(0V - 1.3V) Low (1.7V - 5V) High	(Default) Pulldown to 0V <i>High Signal</i> will cause strike <i>Low Signal</i> will cause retract



Physical And Mechanical Data



Dimensions:

Height	~58mm
Diameter	~40mm

Weight:

About 42 grams (fully assembled)

Mounting Options:

Magnetic Feet

Materials:

ARAM-MOTCA_0_10-OSPG : Pet-G

ARAM-MOTCA_0_10-OSPA : PLA





Operating Conditions & Limitations

Maximum Continuous Temperature : 49 degrees celsius

*Prolonged temperatures above this range will cause armature housing to deform, especially under weight

Environmental Notes : Not guaranteed to be dust or water resistant

Reverse Polarity : failing to follow pin-out and switching the power supply polarity will damage the unit, doing so for a long enough time, with a large enough power supply can be a potential fire hazard.

Magnetic Interference : This device produces a magnetic field, be aware of any damage that may be caused to sensitive electronics, pace makers, etc.

Troubleshooting:

The actuator works most of the time, but when I place it in this specific spot it stops working:

The actuator is sensitive to magnetic interference, metal extremely close to the armature tube can cause a loss in power of the device. Either move the device away from interfering metal, or increase the supply voltage.

Additional Resources:

https://github.com/ARAM-USA-Support/ARAM-MOTCA_0_10-0S

(MOTCA-1 Toggle Actuator Github Documentation Mirror)

*All documentation is also available at www.ARAM-USA.com

For any additional troubleshooting or questions please:

View Our Website : www.ARAM-USA.com

Contact Us At: support@ARAM-USA.com



ARAM-MOTCA_0_10-OSP* Spec Sheet

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Sincerely:

Jack Serlin,

Founder of American Robotics Assisted Manufacturing

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